POWER BI PROJECT REPORT

Credit Card Transactions Analysis

Interactive Business Intelligence Dashboard using Microsoft Power BI

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Platform	Microsoft Power BI
Project file	credit_card_customer_transaction_analysis.pbix
Edition	GitHub Portfolio

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1. Abstract

This project presents an interactive Power BI report for exploring credit-card customer profiles, transaction activity, spending patterns, and selected credit-risk indicators. The supplied PBIX contains two report pages: Customer Overview and Transaction Report.

The report combines summary KPI cards, demographic breakdowns, card-category comparisons, monthly trends, expense-category analysis, payment-method comparisons, and customer segmentation visuals. Interactive slicers allow users to explore the report by date, card type, education, gender, age group, income band, and expense category.

The purpose of this portfolio project is to demonstrate practical skills in data visualization, business intelligence, analytical storytelling, and the presentation of customer and transaction data.

2. Introduction

Credit-card transaction data can help organizations understand customer behavior, monitor spending patterns, compare product segments, and review indicators associated with repayment risk. A well-designed dashboard makes these patterns easier to explore than static tables alone.

This project uses Microsoft Power BI to organize customer and transaction measures into two connected analytical views. The Customer Overview page focuses on customer composition, card products, income, credit limits, satisfaction, and delinquency indicators. The Transaction Report page focuses on transaction value and volume, growth, spending categories, payment methods, and customer-level segmentation.

3. Project Objectives

• Summarize customer and transaction performance through KPI cards.

• Compare customers and credit limits across card categories and demographic groups.

• Explore transaction value and volume over time.

• Analyze spending across expense categories and payment methods.

• Provide filters that support interactive exploration of the report.

• Present selected delinquency and risk indicators alongside customer and spending measures.

4. Tools and Technologies

Microsoft Power BI Desktop — report development and interactive visualizations.

Power Query — data preparation and transformation capabilities within the Power BI workflow.

DAX — measures and calculated logic used for analytical KPIs, where defined in the model.

PBIX — Power BI project file format used to package the report and its model.

5. Report Structure and Dashboard Pages

5.1 Customer Overview

This page presents customer-focused analysis. The visible report elements include total customers, total revenue, total annual fees, average satisfaction score, delinquency percentage, and related status or delta labels. Charts compare customer counts by age group and card category, average credit limit by card type, customers by education level, average monthly income by gender, average credit limit by income band, and delinquency rate by age group. A customer summary table combines education level, income group, revenue, delinquency, average credit limit, and risk level.

5.2 Transaction Report

This page presents transaction-focused analysis. Its KPI cards include total transaction amount, total transaction volume, transaction growth percentage, and total interest earned. Visuals show monthly transaction amount, average

transaction value by card category, spending by expense category, payment-method share (chip, swipe, and online), and customer spend-versus-risk segmentation. A monthly summary table includes revenue month-over-month percentage, transaction volume, transaction growth, and transaction amount.

6. Interactive Filters

The report includes slicers for date (including year, quarter, and month), card category, education level, gender, age group, income group, and expense type. These filters support focused comparisons and help users investigate a selected segment without changing the underlying report design.

7. Key Measures and Analytical Concepts

The visual fields in the PBIX refer to measures including Total Customers, Total Revenue, Total Annual Fees, Avg Satisfaction Score, Delinquent %, Avg Credit Limit, Avg Monthly Income, Total Transaction Amount, Total Transactions Volume, Transaction Growth %, Revenue MoM %, Avg Transaction Value, Contribution %, and Total Interest Earned.

These measures support common business questions such as: Which customer groups contribute to the customer base? How does transaction value vary by month or card category? Which expense categories account for a larger share of spending? How do spending and delinquency indicators vary across income or age segments?

Note: This report documents the measures and visuals identified in the supplied PBIX. Exact DAX formulas and numeric KPI results are not reproduced here because the model's calculation definitions and refreshed data values were not independently extracted and validated.

8. Dashboard Observations

The Customer Overview page is designed to compare demographic and product segments, with dedicated visuals for age, education, gender, income band, and card category.

The Transaction Report page provides both time-based and categorical views of spending, including monthly movement, expense-category contribution, card-category averages, and payment-method distribution.

The report contains explanatory annotations such as a note about the 55+ age cohort and delinquency threshold, a graduate-segment customer share, and comparisons between high- and low-income credit limits. These are dashboard annotations and should be revalidated against refreshed data before being presented as definitive findings.

Because the supplied PBIX was inspected at report-definition level, this document does not claim specific current totals, percentages, or trends beyond the labels and annotations present in the report.

9. Data Preparation and Modeling

A typical workflow for this type of analysis includes importing customer and transaction tables, checking data types, handling missing or unknown category values, validating date fields, and ensuring that customer and transaction records are related correctly.

The report references a CreditCard table, a customer table, and a DateTable. The exact relationship cardinalities, source-system provenance, row counts, and detailed Power Query steps should be confirmed in Power BI Model view and Power Query Editor before final academic or business publication.

10. Conclusion

The Credit Card Transactions Analysis dashboard brings customer profile analysis and transaction analysis into one interactive Power BI report. Its two-page structure helps users move from understanding who the customers are to examining how transactions are distributed over time, categories, and payment methods.

The project demonstrates the use of KPI cards, slicers, trend charts, categorical comparisons, summary tables, and segmentation visuals to support exploratory business analysis.

11. Future Scope

• Add a documented data dictionary and source-data provenance.

• Include validated DAX definitions and a measure glossary.

• Add drill-through pages for customer or transaction segment details, where privacy and data permissions allow.

• Add anomaly detection or forecast views for transaction trends.

• Validate risk thresholds with domain stakeholders and document the methodology.

• Publish a sanitized sample dataset and dashboard screenshots for reproducibility.

12. How to Use the Project

1. Download the PBIX file from the GitHub repository.

2. Open it in Microsoft Power BI Desktop.

3. Allow the report to load and refresh only when the configured data source is available and authorized.

4. Use the slicers to filter the Customer Overview and Transaction Report pages.

5. Hover over visuals and use available interactions to explore segment-level details.

13. Portfolio and Data Disclaimer

This project is presented for learning and portfolio demonstration. Do not upload real cardholder data, personally identifiable information, confidential transaction records, credentials, or proprietary source files to a public repository. Use only data that you own or are authorized to publish. If the PBIX contains sensitive or licensed data, keep the repository private or replace the data with a sanitized sample.